

CHERRYL MARIA BIBIN

Software Engineer | Software Engineer | Cloud-Native & Distributed Systems | Generative AI



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SUMMARY

Software Engineer with 3+ years of experience building and operating backend services, distributed data pipelines, and event-driven systems for enterprise engineering platforms. Experienced in Go, Python, gRPC, NATS, OpenSearch, Cassandra, PostgreSQL, Redis, Kubernetes, and AWS, with hands-on ownership across data ingestion, analytics, integrations, observability, and production reliability. Previously developed LLM-powered applications and RAG pipelines using Python, LangChain, Weaviate, GPT, and LLaMA. Strong experience collaborating across product, SRE, QE, security, and platform teams to deliver scalable backend systems and resolve production issues.

TECHNICAL SKILLS

Languages: Golang, Python, JavaScript, SQL, C++

Backend & Distributed Systems: Microservices, gRPC, REST APIs, Event-Driven Architecture, Distributed Systems, NATS, API Integration

Data & Analytics: OpenSearch, Cassandra, PostgreSQL, Redis, MongoDB, Data Pipelines, Data Modeling

Generative AI / ML: Claude Code, GitHub Copilot, Large Language Models (GPT, LLaMA), LangChain, Retrieval-Augmented Generation (RAG), Vector Embeddings, Weaviate, Prompt Engineering

Cloud & DevOps: Kubernetes, Docker, AWS, AWS S3, Google Cloud Platform, GitHub Actions, CI/CD, Linux

Observability & Developer Tools: Datadog, Git, GitHub, Bazel, Poetry

Frontend: ReactJS, React Native

PROFESSIONAL EXPERIENCES

Company: [CloudBees](#)

Post: Software Engineer

Duration: 2.5 years, Bangalore (ongoing)

Data Platform & Distributed Systems

- Developed and maintained **Go-based gRPC microservices and event-driven data pipelines** powering enterprise Value Stream Management platforms across CI/CD, SCM, security, Jira, testing, and automation domains.
- Worked extensively on the platform's **data ingestion architecture**, where NATS events are routed through specialized processing pipelines, transformed into internal models, and indexed into OpenSearch for downstream analytics.
- Implemented and debugged ingestion workflows involving **NATS event routing, OpenSearch indexing, Redis caching, and distributed service dependencies**, addressing data consistency, cache invalidation, and pipeline failures.
- Worked on Cassandra-backed project management services responsible for synchronizing Jira projects, issues, and sprints, including **rate limiting, distributed locks, polling workflows, and API synchronization**.
- Improved production reliability through **Datadog tracing, application performance monitoring, production debugging, and monitoring instrumentation**, tracing ingestion and API issues across distributed services.

Analytics & VSM Platform

- Developed backend functionality for **Reports and Analytics services**, powering enterprise dashboards across DORA, Flow, CI Insights, Security Insights, Test Insights, and software delivery metrics.
- Designed and implemented **OpenSearch queries, widget payloads, transformation logic, and backend APIs** used to calculate and serve engineering analytics to VSM dashboards.
- Worked across **40+ OpenSearch indices and multiple data domains**, maintaining index lifecycle operations, mappings, schemas, ingestion compatibility, and downstream analytics requirements.
- Optimized backend APIs, data processing paths, and query execution to improve **dashboard responsiveness and analytics performance across high-traffic services**.
- Implemented test-result processing workflows including **JUnit/Jenkins result parsing, S3-based data retrieval, chunked processing, and NATS-based downstream publishing**.

Integrations, Security & Production Engineering

- Delivered integrations across **Jenkins, GitHub Actions, GitLab, Gerrit, and Jira**, implementing event flows and backend functionality to surface build, SCM, security, testing, and project-management data in enterprise dashboards.
- Owned backend development for **application security dashboards**, including **remediation and SLA status events**, widget payloads, transformations, unit tests, and GitHub Actions data integration.
- Investigated and resolved production issues involving **ingestion bottlenecks, pod restarts, cache failures, API load, rate limiting, and analytics visibility**, helping unblock enterprise customers and restore platform functionality.
- Implemented monitoring and metrics endpoints for backend services, improving visibility for SRE teams and accelerating diagnosis of production performance and reliability issues.
- Collaborated with **SRE, QE, ASPM, Core, Product, and external platform teams** across feature delivery, production incidents, testing, integrations, customer issues, and technical troubleshooting.
- Took ownership of critical deliverables during team capacity gaps, including **Jenkins integrations, security analytics, test-result processing, and production issue resolution**.

Technologies: Go, gRPC, NATS, OpenSearch, Cassandra, PostgreSQL, Redis, Kubernetes, Docker, Datadog, GitHub Actions, Jenkins, AWS S3

Company: [Machani Robotics](#)

Post: Software Engineer

Duration: 6 months(Bangalore)

- Designed and developed **LLM-powered conversational AI applications** using GPT and LLaMA, enabling contextual natural-language interactions for humanoid robotics.
- Built Retrieval-Augmented Generation (RAG)** pipelines using LangChain, **vector embeddings**, and Weaviate to improve contextual knowledge retrieval for AI-driven interactions.
- Developed backend services and API integrations using Python and Go, incorporating **Redis caching, PostgreSQL, and vector databases** to support low-latency AI inference.
- Implemented event-driven communication between AI services using **Redis Pub/Sub, gRPC, and Cap'n Proto**, improving service orchestration and interoperability.
- Collaborated on technical demonstrations and customer-facing AI prototypes for organizations and research institutions including **NVIDIA and UC San Diego**.

Technologies: Python, Go, LangChain, Weaviate, PostgreSQL, Redis, gRPC, Cap'n Proto, Poetry, Bazel

Company: [Meta \(Facebook\)](#) **Post:** Data Center Engineering Intern **Duration:** 3 months (North Carolina, USA)

- Developed Python-based automation and analyzed production-scale infrastructure data to support **server fleet reliability, operational efficiency, and production engineering workflows**.
- Built core functionality for internal automation tools used for **testing, validation, and break-fix workflows**, improving operational efficiency for production engineers.
- Queried and analyzed infrastructure data using **SQL and Presto** to identify operational patterns and opportunities for automation.

Technologies: Python, SQL, Presto, Linux

Company: [Logituit](#) **Post:** Software Development Intern **Duration:** 3 months (Bangalore)

- Developed an **OTT analytics dashboard** for collecting and analyzing user-experience data across streaming platforms.
- Built REST APIs and backend functionality using Node.js and developed dashboard interfaces using **AngularJS, integrating MongoDB for data storage**.

Technologies: Node.js, MongoDB, AngularJS, REST APIs

Company: [The University of Texas at Arlington](#) **Post:** Undergraduate Researcher **Duration:** 3 years (Arlington, Texas, USA)

- Conducted undergraduate research involving **human-computer interaction, biosignal data**, and Python-based data analysis through the **Undergraduate Research Opportunity Program**.
- Developed prototypes involving **NLP chatbots, computer vision with OpenCV**, and interactive web applications using React and Node.js.
- Built real-time **interactive systems using WebSockets**, React, Node.js, Python, and MongoDB.

Technologies: Python, ReactJS, Node.js, OpenCV, MongoDB, WebSockets

TECHNICAL PROJECTS

Project: Interactive Living Stream **Team Size:** 5 **Duration:** 4 months

- Collaborated with the **U.S. Army Corps of Engineers (USACE)** to develop an interactive augmented-reality experience for national park visitors.
- Built a projection-based interactive environment using **Unity, C#, motion sensors, and Intel RealSense**, enabling projected fish to respond dynamically to visitor movement.

Technologies: Unity, C#, Intel RealSense, Motion Sensors | **Project Link:** [🔗](#)

Project: The Battles Within **Team Size:** 5 **Duration:** 3 months

- Developed a **Unity-based Android game** focused on mental-health awareness, implementing multiple game modes, player metrics, animations, and interactive gameplay systems.
- Designed application logic, data storage, and user-facing game interactions using **C# and Unity**.

Technologies: Unity, C#, Adobe Illustrator, Adobe After Effects | **Project Link:** [🔗](#)

Project: AirMon **Team Size:** 5 **Duration:** 1 week

- Developed an IoT system for real-time **air-quality and location monitoring using Arduino**, environmental sensors, GPS, and cloud-based visualization.
- Integrated sensor data with **Google Maps and ThingSpeak for geographic visualization** and analysis.

Technologies: C++, Arduino, ThingSpeak, Google Maps | **Project Link:** [🔗](#)

ACHIEVEMENTS AND EVENTS

- Maverick Academic Scholarship**, The University of Texas at Arlington — Recipient for **four** consecutive years (2019–2023).
- UROP Award**, The University of Texas at Arlington — Undergraduate research funding and faculty mentorship.
- 1st Place, WeTech** Regional Pitch — Bangalore, India.
- Global Finalist, WeTech** World Pitch — San Francisco, USA.
- Regional Finalist, Schneider Electric** Go Green 2021 Challenge — North America.
- Participant, Infosys** Summer of Ideas Challenge.

ACADEMICS

The University of Texas at Arlington — Arlington, Texas, USA

B.Tech., Computer Science Engineering | **GPA: 3.6/4.0** (2019–2023)

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, Artificial Intelligence, Machine Learning, Cloud Computing